

Worksheet 3

Math 1A

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Problem Q1

Can the graph of a function have more than one tangent at a given point? If so, graph your answer. If not, explain why.

Solution.

对函数图像 $y = f(x)$, 若 $f'(a)$ 存在, 则切线唯一。

一般曲线 (非函数图像) 在自交点处可以有多条切线。

Final Answer:

Problem Q2

Is there a function whose graph doesn't have a tangent at some point? If so, graph your answer. If not, explain why.

Solution.

存在。例如 $f(x) = |x|$ 在 $x = 0$ 处有尖角, 不可导, 没有切线。

其它例子: $x^{1/3}$ 在 $x = 0$ 处有垂直切线。

Final Answer:

$|x| \ x = 0$

Problem P1

Suppose δ is a positive real number. How do you describe all real numbers x that are within δ of 0?

Solution.

$x = x$

Final Answer:

x

Problem P2

Suppose a is a real number and δ is a positive real number. How do you describe all real numbers x that are within δ of a ?

Solution.

$a = a$

Final Answer:

$$a$$

Problem P3

Suppose a is a real number and δ is a positive real number. How do you describe all real numbers x that are within δ of a , but not equal to a :

Solution.

$$a = a$$

Final Answer:

$$a$$

Problem P4

$$\text{Let } f(x) = x^2.$$

Solution.**Final Answer:**

见子题

(a) $x = x$ x

(b) $x = x$ x

(c) $x = x$ x

(d) $fx = fx$ fx

Problem AP1

Graph $y = 2x - x^2$. For which points a is the tangent line to the curve at the point $(a, 2a - a^2)$ a horizontal line? How is the value of $2x - x^2$ at these points related to the values of $2x - x^2$ at other points?

Solution.

$$f(x) = -x^2 + 2x$$

$$f'(x) = 2 - 2x$$

$$\text{令 } f'(x) = 0: x = [1]$$

Final Answer:

$$2 - 2x$$

Problem AP2

The point $(2, 0)$ is on the curve $y = 2x - x^2$.

Solution.

$$f(x) = -x^2 + 2x$$

$$f'(x) = 2 - 2x$$

Final Answer:

$2 - 2x$

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